The Complete Treatise on the Two-Institutions Model

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Abstract

This treatise develops a formal framework for two institutions that assess a common object, asset, policy, obligation, or economic state. Each institution has an irreducible intrinsic-value benchmark, a self-valuation, and a dual valuation assigned by the other institution. The purpose is not to replace disagreement with consensus at any cost, but to reduce disagreement while preserving accuracy, accounting transparency, institutional independence, and lawful constraints. The framework combines a disjunctive multiplicative condition, a loss-based reconciliation objective, dynamic adjustment, value-based accounting, monetary-asset market equilibrium, relative-price consistency, econometric identification, risk analysis, and an implementation protocol. It distinguishes model-implied intrinsic value from market price, utility, institutional belief, and legal entitlement. Cross-disciplinary extensions are included where they clarify measurement, governance, welfare, security, and the limits of inference. The resulting system is a research architecture rather than a claim that every valuation problem has a uniquely observable intrinsic value.

The treatise ends with "The End"

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1 Purpose, scope, and epistemic discipline

The Two-Institutions Model studies two institutions, indexed by $i \in \{1, 2\}$, that evaluate a common object or a set of economically related objects. Examples include monetary assets, sovereign obligations, collateral, public projects, regulatory decisions, financial claims, and institutional performance.

The central problem is that an institution's own assessment may differ from the assessment assigned to it by the other institution. The framework therefore separates three quantities that must not be collapsed into one:

1. an irreducible intrinsic-value benchmark;
2. an institution's self-valuation;
3. a dual valuation assigned by the other institution.

The word "intrinsic" is a modeling designation. It does not imply that the benchmark is metaphysically observable, institution-free, or empirically identified by definition. The benchmark must be operationalized through an explicit structural model, independent evidence, or a justified measurement convention.

The theory has five objectives:

1. represent institutional valuations without confusing them with market prices;
2. measure disagreement and valuation error separately;
3. constrain reconciliation so that agreement is not mistaken for truth;
4. account separately for changes in fundamentals, beliefs, and cross-institution assessments;
5. use economic equilibrium and external evidence to discipline, but not mechanically dictate, institutional valuations.

The treatise distinguishes three levels of claim. A *definition* establishes notation. An *assumption* specifies a modeling restriction. A *proposition* follows mathematically from stated assumptions. Empirical and legal conclusions require evidence beyond algebra.

2 Notation and the primitive valuation structure

Let $t \in [0, T]$ denote time. For each institution i, define

$$V_i^*(t) \equiv \text{irreducible intrinsic-value benchmark}, \tag{1}$$

$$S_i(t) \equiv \text{self-valuation of institution } i, \tag{2}$$

$$D_i(t) \equiv \text{dual valuation assigned to institution } i \text{ by institution } j \neq i. \tag{3}$$

Thus D_1 is institution 2's valuation of the object or position associated with institution 1, and D_2 is institution 1's corresponding dual valuation. The term "dual" denotes the cross-institution assessment; it does not automatically mean a mathematical dual space or a financial derivative.

The institution-specific valuation vector is

$$\mathbf{v}_i(t) = \begin{pmatrix} V_i^*(t) \\ S_i(t) \\ D_i(t) \end{pmatrix}, \tag{4}$$

and the complete state vector is

$$\mathbf{z}(t) = \begin{pmatrix} V_1^*(t) & S_1(t) & D_1(t) & V_2^*(t) & S_2(t) & D_2(t) \end{pmatrix}^{\mathsf{T}}. \tag{5}$$

2.1 Irreducibility

The benchmark V_i^* is not defined as the arithmetic mean of S_i and D_i. In particular, the identity

$$V_i^* \neq \frac{S_i + D_i}{2} \tag{6}$$

is a methodological prohibition on defining the benchmark by consensus alone, not a claim that equality can never occur numerically. The benchmark may be estimated from cash flows, scarcity, service flows, risk, legal rights, physical constraints, or a structural equilibrium model. Its status depends on the object being valued.

2.2 Errors and disagreement

Define self-valuation error and dual-valuation error by

$$e_i^S(t) = S_i(t) - V_i^*(t), \tag{7}$$

$$e_i^D(t) = D_i(t) - V_i^*(t). \tag{8}$$

Define signed disagreement as

$$\Delta_i(t) = S_i(t) - D_i(t). \tag{9}$$

Then

$$\Delta_i = e_i^S - e_i^D. \tag{10}$$

This identity is central: disagreement is the difference between two errors, not a direct measure of either error's magnitude.

Proposition 1 (Consensus need not imply accuracy)**.** *For any benchmark V_i^*, the condition $S_i = D_i$ implies $\Delta_i = 0$, but does not imply $S_i = V_i^*$ or $D_i = V_i^*$.*

Proof. Take $V_i^* = 100$ and $S_i = D_i = 120$. Then $\Delta_i = 0$, while $e_i^S = e_i^D = 20$. Thus disagreement is zero and both assessments are inaccurate relative to the benchmark. □

3 The multiplicative ideal condition

The proposed institution-specific multiplicative condition is

$$\Phi_i(t) = \big(S_i(t) - D_i(t)\big)\big(D_i(t) - V_i^*(t)\big)\big(V_i^*(t) - S_i(t)\big) = 0, \qquad i \in \{1, 2\}. \tag{11}$$

The condition is imposed separately for each institution. It is a disjunctive constraint: at least one factor must vanish.

3.1 Branch interpretation

Equation (11) is equivalent to the union of three branches:

$$\mathcal{B}_{SD,i}: \quad S_i = D_i, \tag{12}$$

$$\mathcal{B}_{DV,i}: \quad D_i = V_i^*, \tag{13}$$

$$\mathcal{B}_{VS,i}: \quad V_i^* = S_i. \tag{14}$$

It does not, by itself, require all three values to coincide. The fully reconciled state is the intersection

$$S_i = D_i = V_i^*. \tag{15}$$

Proposition 2 (Logical meaning of the multiplicative condition)**.** *For real-valued S_i, D_i, V_i^*, equation (11) holds if and only if at least two of the three values S_i, D_i, V_i^* are equal.*

Proof. A product of real numbers is zero if and only if at least one factor is zero. Each factor in (11) is a pairwise difference. Therefore at least one pair is equal. Conversely, if any pair is equal, the corresponding factor is zero and the product vanishes. □

3.2 Logical caution

The condition is not a metric, a ranking, a causal mechanism, or a proof of accuracy. It does not tell us which branch is normatively preferred. In particular, the branch $S_i = D_i$ may describe consensus around a false assessment. A loss function that separately rewards accuracy is needed to prefer full reconciliation when the benchmark is known and the feasible set permits it.

The condition also does not require $V_1^* = V_2^*$. If the two benchmarks refer to distinct objects, equality may be inappropriate. If they refer to the same object measured under the same convention, a cross-institution consistency restriction may be justified, but it must be stated separately.

4 Loss functions: disagreement, accuracy, and reconciliation

Let $w_i > 0$ be disagreement weights, and let $a_i, b_i > 0$ be accuracy weights. Define

$$\mathcal{D}(\mathbf{z}) = \sum_{i=1}^{2} w_i (S_i - D_i)^2, \tag{16}$$

$$\mathcal{A}(\mathbf{z}) = \sum_{i=1}^{2} \left[a_i (S_i - V_i^*)^2 + b_i (D_i - V_i^*)^2 \right]. \tag{17}$$

The combined objective is

$$\mathcal{J}(\mathbf{z}) = \lambda \mathcal{D}(\mathbf{z}) + (1 - \lambda)\mathcal{A}(\mathbf{z}), \qquad 0 < \lambda < 1. \tag{18}$$

The parameter λ controls the relative emphasis on institutional agreement versus benchmark accuracy. It should be selected transparently and subjected to sensitivity analysis. Since $\mathcal{D}$ and $\mathcal{A}$ may have different units or scales, practical estimation may require standardization or economically justified weights.

Proposition 3 (Global reconciliation under an observed benchmark)**.** *Suppose V_i^* is fixed and known, all four accuracy weights are strictly positive, and the feasible set permits $S_i = D_i = V_i^*$. Then the global minimum of $\mathcal{J}$ is zero and is attained at $S_i = D_i = V_i^*$ for each institution. This conclusion holds for every $\lambda \in (0, 1)$.*

Proof. Every term in $\mathcal{D}$ and $\mathcal{A}$ is nonnegative. Hence $\mathcal{J} \geq 0$. At $S_i = D_i = V_i^*$, all terms vanish, so $\mathcal{J} = 0$. Since $1 - \lambda > 0$ and $a_i, b_i > 0$, a zero objective requires each self- and dual-valuation error to be zero. Thus the minimizer is the fully reconciled state. □

This proposition is conditional, not a universal empirical result. If the benchmark is latent, estimated with error, or constrained by legal mandates, accounting rules, capital limits, political objectives, or asymmetric information, the minimizer may not be feasible or uniquely identified.

4.1 Constrained reconciliation

A general reconciliation problem can be written as

$$\min_{\mathbf{z},\theta} \quad \mathcal{J}(\mathbf{z}, \theta) + \mu \mathcal{L}_{\text{market}}(\mathbf{z}, \theta) + \nu \mathcal{R}(\theta) \tag{19}$$

$$\text{subject to} \quad \Phi_i(\mathbf{z}) = 0, \quad i \in \{1, 2\}, \tag{20}$$

$$\mathbf{z} \in \mathcal{F}_{\text{legal}} \cap \mathcal{F}_{\text{accounting}} \cap \mathcal{F}_{\text{physical}}, \tag{21}$$

$$Q_j^D(P, X; \theta) = Q_j^S(P, Z; \theta), \quad j \in M. \tag{22}$$

Here θ denotes structural parameters; $\mathcal{L}_{\text{market}}$ penalizes inconsistency with observed market relations; $\mathcal{R}$ regularizes parameters; and the feasible sets encode applicable constraints. The coefficients μ, ν must be selected with attention to units, identification, and the consequences of misspecification.

5 Value-based accounting

Value accounting separates changes in the benchmark from changes in the institution's assessments. For each institution,

$$\mathrm{d}e_i^S = \mathrm{d}S_i - \mathrm{d}V_i^*, \tag{23}$$

$$\mathrm{d}e_i^D = \mathrm{d}D_i - \mathrm{d}V_i^*, \tag{24}$$

$$\mathrm{d}\Delta_i = \mathrm{d}S_i - \mathrm{d}D_i = \mathrm{d}e_i^S - \mathrm{d}e_i^D. \tag{25}$$

If both assessments are benchmarked against the same moving intrinsic benchmark, the benchmark change cancels from the difference between them. It does not follow that the benchmark itself is unchanged.

5.1 A change ledger

For a discrete period $t \to t+1$, define

$$\delta V_{i,t}^* = V_{i,t+1}^* - V_{i,t}^*, \tag{26}$$

$$\delta S_{i,t} = S_{i,t+1} - S_{i,t}, \tag{27}$$

$$\delta D_{i,t} = D_{i,t+1} - D_{i,t}. \tag{28}$$

The accounting bridge is

$$\delta e_{i,t}^S = \delta S_{i,t} - \delta V_{i,t}^*, \tag{29}$$

$$\delta e_{i,t}^D = \delta D_{i,t} - \delta V_{i,t}^*, \tag{30}$$

$$\delta \Delta_{i,t} = \delta S_{i,t} - \delta D_{i,t}. \tag{31}$$

An auditable ledger should record the source, timestamp, unit, valuation convention, model version, responsible party, and evidence for each revision. This is an accounting architecture, not a substitute for applicable statutory accounting standards.

Table 1: Minimum value-accounting ledger

Entry	Quantity	Required interpretation
Benchmark revision	δV_i^*	Change in the structural or independently measured benchmark
Self revision	δS_i	Change in the institution's own assessment
Dual revision	δD_i	Change in the other institution's assessment
Self error change	δe_i^S	Self revision net of benchmark revision
Dual error change	δe_i^D	Dual revision net of benchmark revision
Disagreement change	$\delta \Delta_i$	Difference between self and dual revisions

6 Dynamic adjustment and stability

A possible continuous-time adjustment law is

$$\dot{S}_i = \dot{V}_i^* - \alpha_i e_i^S - \beta_i \Delta_i + u_i^S, \tag{32}$$

$$\dot{D}_i = \dot{V}_i^* - \gamma_i e_i^D + \delta_i \Delta_i + u_i^D, \tag{33}$$

where α_i, γ_i govern error correction, β_i, δ_i govern disagreement adjustment, and u_i^S, u_i^D represent disturbances, policy shifts, information arrivals, or model errors. These are proposed laws, not empirically established behavioral laws.

Subtracting the two equations gives

$$\dot{\Delta}_i = -\alpha_i e_i^S + \gamma_i e_i^D - (\beta_i + \delta_i)\Delta_i + u_i^S - u_i^D. \tag{34}$$

If the error-driven and disturbance terms are zero, then

$$\dot{\Delta}_i = -(\beta_i + \delta_i)\Delta_i. \tag{35}$$

Proposition 4 (Conditional exponential decay)**.** *If $\beta_i + \delta_i > 0$ and the forcing term $-\alpha_i e_i^S + \gamma_i e_i^D + u_i^S - u_i^D$ is identically zero, then*

$$\Delta_i(t) = \Delta_i(0) \exp\left(-(\beta_i + \delta_i)t\right). \tag{36}$$

Consequently, $\Delta_i(t) \to 0$ as $t \to \infty$.

Proof. The differential equation is linear and separable. For nonzero Δ_i, divide by Δ_i, integrate, and exponentiate. The resulting formula also holds when the initial disagreement is zero. □

This result establishes convergence of disagreement under the specified conditions only. It does not establish convergence of either assessment to V_i^*. Forcing terms, delayed information, strategic behavior, measurement error, or changing benchmarks may prevent convergence or produce oscillation.

6.1 Preserving the multiplicative constraint

For differentiable trajectories, the derivative of the constraint is

$$\begin{aligned}\dot{\Phi}_i = &(\dot{S}_i - \dot{D}_i)(D_i - V_i^*)(V_i^* - S_i) \\ &+ (S_i - D_i)(\dot{D}_i - \dot{V}_i^*)(V_i^* - S_i) \\ &+ (S_i - D_i)(D_i - V_i^*)(\dot{V}_i^* - \dot{S}_i).\end{aligned} \tag{37}$$

A trajectory that remains on the constraint set must satisfy $\dot{\Phi}_i = 0$. At singular intersections of branches, this tangency condition alone may not guarantee that a chosen numerical integrator preserves the constraint. A practical implementation can use a constrained update, branch-aware solver, or projection step after each update.

7 A bounded reciprocal valuation option

An optional restriction inspired by the bounded-dual-valuation framework is available when $V_i^* > 0$. Define normalized assessments

$$F_i = \frac{S_i}{V_i^*}, \qquad G_i = \frac{D_i}{V_i^*}. \tag{38}$$

One may impose

$$F_i = e^{h_i}, \qquad G_i = e^{-h_i}, \tag{39}$$

which implies

$$F_i G_i = 1, \tag{40}$$

$$S_i D_i = (V_i^*)^2, \tag{41}$$

$$S_i + D_i = 2V_i^* \cosh(h_i), \tag{42}$$

$$S_i - D_i = 2V_i^* \sinh(h_i). \tag{43}$$

This construction guarantees positive S_i, D_i when $V_i^* > 0$, and reciprocal normalized deviations. It is a substantive modeling axiom, not a logical consequence of having two institutions. It is inappropriate without modification when values may be zero or negative, or when multiplicative reciprocity has no economic interpretation.

8 Monetary assets: demand, supply, and relative prices

The monetary-asset framework supplies an external economic structure with which institutional assessments may be tested. It treats monetary assets as competing assets and emphasizes relative prices, asset-specific demand, scarcity, network effects, institutional credibility, liquidity, collateral utility, and geopolitical conditions. A dollar quote is a relative price; it is not automatically an ontologically neutral measurement of the quoted asset's intrinsic value.

Let M be a set of monetary assets. A generic asset-attractiveness or utility index for asset j can be represented as

$$U_j = U_j(R_j, \sigma_j, L_j, T_j, N_j, S_j, I_j, K_j, G_j, X), \tag{44}$$

where R_j is expected return, σ_j risk, L_j liquidity, T_j transaction utility, N_j network utility, S_j scarcity, I_j institutional credibility, K_j collateral utility, G_j geopolitical characteristics, and X other state variables. This notation is consistent with the uploaded monetary-asset treatise; the exact functional form must be estimated or justified.

A convenient portfolio-share specification is

$$s_j = \frac{\exp(U_j)}{\sum_{k \in M} \exp(U_k)}, \qquad \sum_{j \in M} s_j = 1. \tag{45}$$

This softmax form is a modeling choice. It yields nonnegative shares summing to one, but it does not by itself establish causal interpretation or a unique structural utility scale.

For assets i, j, define the relative price

$$P_{ij} = \frac{P_i}{P_j}. \tag{46}$$

For positive prices, relative-price consistency requires

$$P_{ij} P_{jk} = P_{ik}, \tag{47}$$

$$P_{ij} = \frac{1}{P_{ji}}, \tag{48}$$

$$P_{ii} = 1. \tag{49}$$

These identities are algebraic consistency conditions. Observed violations can reflect asynchronous timestamps, bid–ask spreads, transaction costs, market segmentation, or data errors, rather than an economically exploitable arbitrage opportunity.

8.1 Demand–supply equilibrium

For each monetary asset $j \in M$, let

$$Q_j^D = D_j(P, X; \theta_D), \tag{50}$$

$$Q_j^S = S_j(P, Z; \theta_S), \tag{51}$$

where P denotes a normalized system of relative prices, X demand-side state variables, Z supply-side state variables, and θ_D, θ_S structural parameters. Market clearing is

$$D_j(P, X; \theta_D) = S_j(P, Z; \theta_S), \qquad j \in M. \tag{52}$$

Because only relative prices are economically meaningful in this representation, one price can be normalized mathematically. Such normalization is not a claim that the numeraire is economically or philosophically privileged.

Proposition 5 (Market clearing does not identify intrinsic value by itself)**.** *Suppose observed equilibrium prices and quantities satisfy* (52)*. Without additional restrictions on preferences, technologies, expectations, institutions, and the mapping from fundamentals to valuation, the equilibrium condition alone does not uniquely identify* V_j^**.*

Proof. Market clearing imposes equality between demand and supply at equilibrium. Distinct demand and supply functions can intersect at the same observed price and quantity while implying different counterfactuals and different structural valuations. Therefore the equilibrium equality, without identifying restrictions or additional evidence, does not uniquely determine an intrinsic-value benchmark. □

8.2 Connecting market equilibrium to dual valuation

The monetary equilibrium model can inform the benchmark, but must not be confused with it. A useful architecture distinguishes:

1. *market price*: an observed exchange ratio at a time and venue;
2. *utility or attractiveness index*: a model-dependent representation of asset services and preferences;
3. *intrinsic benchmark*: a structural or independently measured reference value;
4. *self-valuation*: an institution's assessment;
5. *dual valuation*: the other institution's assessment.

A benchmark may be estimated as $V_j^* = g_j(X, Z; \theta)$, but the function g_j must be justified and identified. A market-clearing price is one input to this exercise, not a proof that the price equals intrinsic value.

9 Econometric identification and empirical design

The model's empirical component must separate measurement from inference. Let Y_t denote observed data, including prices, quantities, yields, spreads, liquidity measures, asset supplies, transaction volumes, network activity, policy variables, and institution-level assessments. Let θ collect structural parameters. A generic measurement system is

$$Y_t = h(\theta, X_t) + \varepsilon_t, \tag{53}$$

$$V_{i,t}^* = g_i(\theta, X_t) + \eta_{i,t}, \tag{54}$$

where ε_t and $\eta_{i,t}$ represent measurement and specification errors. These errors may be correlated and heteroskedastic; assumptions must be stated rather than presumed.

9.1 Identification requirements

Point identification requires that the available observations and restrictions distinguish the target parameter or benchmark from alternatives. The following checks are necessary:

1. **Scale and normalization:** specify units and any price normalization.
2. **Rank and variation:** ensure the data contain variation capable of separating parameters.
3. **Exclusion restrictions:** justify why an instrument affects the target through the claimed channel rather than through omitted direct channels.

4. **Simultaneity:** recognize that prices, quantities, expectations, and valuations may be jointly determined.

5. **Measurement validity:** document data definitions, revisions, missingness, and timestamp alignment.

6. **Robustness:** vary functional forms, priors, windows, weights, and normalization conventions.

7. **Out-of-sample assessment:** evaluate predictive calibration and stability, not only in-sample fit.

Potential sources of quasi-experimental variation in monetary-asset research may include protocol-driven supply changes, exogenous mining or staking changes, geological supply disruptions, monetary-policy surprises, technology shocks, regulatory announcements, or geopolitical events. Each proposed instrument requires a case-specific exclusion argument. None is automatically valid merely because it is observable or plausibly external.

9.2 Joint estimation objective

Let $\mathcal{L}_{\text{data}}$ be a statistical loss for observed data, $\mathcal{L}_{\text{market}}$ a market-equilibrium residual, and $\mathcal{L}_{\text{valuation}}$ the valuation objective. A joint estimator may solve

$$\begin{aligned} \min_{\theta,\mathbf{z},P} \quad & \mathcal{L}_{\text{data}}(\theta) + \mu\mathcal{L}_{\text{market}}(P,\theta) + \rho\mathcal{L}_{\text{valuation}}(\mathbf{z},\theta) + \nu\mathcal{R}(\theta) \\ \text{subject to} \quad & \Phi_i = 0, \quad i = 1,2, \\ & Q_j^D(P,X;\theta) = Q_j^S(P,Z;\theta), \quad j \in M. \end{aligned} \tag{55}$$

One possible market loss is

$$\mathcal{L}_{\text{market}} = \sum_{j\in M} \omega_j \left[Q_j^D(P,X;\theta) - Q_j^S(P,Z;\theta)\right]^2, \tag{56}$$

with $\omega_j > 0$. The objective is a proposed estimator, not a theorem that the true model has been recovered. In practice, strict equilibrium constraints may be replaced by soft penalties when data are noisy or market frictions are material.

10 Statistical decision theory and uncertainty

A point estimate can conceal uncertainty. If the benchmark is estimated, write

$$\widehat{V}_i^* = V_i^* + \epsilon_i^V, \tag{57}$$

where ϵ_i^V is benchmark-estimation error. The observed self-error $\widehat{e}_i^S = S_i - \widehat{V}_i^*$ then differs from the true error by $-\epsilon_i^V$. Consequently, a small measured error may reflect benchmark error rather than accuracy.

A probabilistic version can assign a posterior distribution

$$p(V_i^*, \theta \mid Y) \propto p(Y \mid V_i^*, \theta) p(V_i^*, \theta). \tag{58}$$

Posterior uncertainty should be propagated into the disagreement and accuracy metrics. For example,

$$\mathbb{E}\left[(S_i - V_i^*)^2 \mid Y\right] \tag{59}$$

is a posterior expected squared error, not simply the squared difference between S_i and a posterior mean unless the relevant variance term is included.

For squared-error loss,

$$\mathbb{E}\left[(S_i - V_i^*)^2 \mid Y\right] = (S_i - \mathbb{E}[V_i^* \mid Y])^2 + \mathrm{Var}(V_i^* \mid Y). \tag{60}$$

This decomposition distinguishes distance from the posterior mean from uncertainty about the benchmark itself.

11 Financial planning and analysis

For an asset, project, or institutional balance sheet, the framework can be embedded in a financial planning and analysis process. The analysis should keep operating assumptions, financing assumptions, risk premia, liquidity, collateral, and valuation adjustments separate.

For a cash-flow asset with uncertain cash flows C_t, a conventional benchmark may be

$$V^* = \sum_{t=1}^{T} \frac{\mathbb{E}[C_t]}{(1+r_t)^t} + \frac{\mathbb{E}[C_{T+1}]}{(r_\infty - g)(1+r_T)^T}, \tag{61}$$

under the restrictive assumptions of a compatible terminal-growth model, $r_\infty > g$, and coherent discount-rate conventions. This formula is an illustrative valuation method, not a universal definition of intrinsic value. Risk adjustment, default, embedded options, taxes, inflation, currency mismatch, and liquidity may require a richer model.

A useful planning decomposition is

$$V^* = V_{\text{operating}} + V_{\text{financing}} + V_{\text{optionality}} - V_{\text{contingent liabilities}}, \tag{62}$$

provided that the components are defined so as not to double-count cash flows or risks. Institutional disagreement can then be decomposed into differences in forecast cash flows, discount rates, scenario probabilities, liquidity assumptions, legal enforceability, and model structure.

Table 2: Financial analysis and valuation-accounting bridge

Driver	Example measurement	Valuation implication
Cash-flow forecast	Revenue, costs, debt service	Changes expected payoff
Discount rate	Risk-free rate, spread, risk premium	Changes present value
Liquidity	Bid–ask spread, depth, turnover	Changes liquidation and transaction costs
Collateral	Haircut, eligibility, margin requirement	Changes funding capacity
Currency exposure	FX sensitivity and mismatch	Changes converted cash flows and risk
Legal enforceability	Priority, covenants, jurisdiction	Changes expected recovery and constraints
Scenario risk	Default, inflation, policy, conflict	Changes probability-weighted outcomes

12 Law, governance, and institutional accountability

The model is not a legal doctrine. Its use in a regulated setting must respect applicable law, fiduciary duties, contract terms, accounting standards, prudential rules, data-protection obligations, and procedural fairness. A mathematical optimum cannot override a legal prohibition or create a right that the governing law does not recognize.

A governance protocol should specify:

1. the legal identity and mandate of each institution;
2. who may originate, review, challenge, and approve a valuation;
3. the evidentiary standard and permissible data sources;
4. conflict-of-interest controls and independent review;
5. audit trails, version control, retention, and correction procedures;
6. appeal or dispute-resolution routes;
7. when confidential information may be shared;
8. how model risk and human overrides are documented.

The dual valuation should not be treated as a veto or an endorsement by default. Its institutional role must be established by statute, contract, policy, or agreed governance. Where decisions affect persons, firms, or public resources, the system should support reasons-giving, contestability, and proportionality.

13 Physics, chemistry, and biological systems: limits and analogies

The framework may be applied to physical or biological systems only after defining a valid measurement model. Physics distinguishes state variables, observables, conservation laws, and measurement uncertainty. The analogy is useful: a valuation model should distinguish a latent state from an observed measurement and should obey relevant physical constraints. But financial value is not a conserved physical quantity merely because an accounting identity is used.

In chemistry, equilibrium conditions are meaningful only relative to specified species, reactions, activities, temperature, pressure, and conservation constraints. This suggests a methodological lesson: a market-equilibrium claim must identify the objects, feasible transformations, and boundary conditions. It does not imply that financial equilibrium follows chemical thermodynamics.

In biology, adaptive systems can maintain distinct local estimates while exchanging signals. This analogy may motivate distributed estimation and robust feedback, but it does not establish that institutions behave like organisms or that a biological mechanism transfers unchanged to financial governance.

13.1 Measurement discipline

For any cross-disciplinary application, specify

$$\text{latent state} \longrightarrow \text{measurement model} \longrightarrow \text{uncertainty model} \longrightarrow \text{decision rule}. \tag{63}$$

The model should not identify a latent state solely with an instrument reading, a proxy, a reported belief, or a consensus statistic.

14 Linguistics, psychology, and psychiatry

Institutional disagreement can arise from different definitions, reference classes, time horizons, incentives, information sets, or loss functions. Linguistic analysis is therefore not cosmetic: the

same word, such as "value," "risk," "solvency," or "scarcity," may encode different operational concepts across institutions.

A protocol should define each variable, unit, reference date, forecast horizon, and decision purpose. If S_i and D_i refer to different constructs, their numerical difference may be meaningless until they are mapped to a common measurement scale.

Psychology can motivate hypotheses about anchoring, overconfidence, loss aversion, motivated reasoning, and group polarization. These should be tested rather than used to diagnose decision-makers from their valuations. Psychiatry is not a tool for labeling institutional disagreement as illness; clinical diagnoses require appropriate clinical evidence and context. The model should explain observable decisions through transparent incentives and information wherever possible, not speculate about mental health.

A useful decomposition is

$$\Delta_i = \Delta_i^{\text{data}} + \Delta_i^{\text{model}} + \Delta_i^{\text{language}} + \Delta_i^{\text{incentive}} + \Delta_i^{\text{judgment}}, \tag{64}$$

under a defined attribution scheme. This is an accounting decomposition only if the components are operationally defined and interaction terms are handled; otherwise it is a conceptual checklist, not an identified equation.

15 Philosophy, religion, and epistemology

The model makes a modest epistemic claim: agreement and truth are distinct concepts. A consensus can be wrong; disagreement can expose model risk; and an independently specified benchmark can itself be uncertain or biased. The "irreducible" benchmark should therefore be understood as irreducible to the two institutions' agreement alone, not necessarily irreducible to all observation, explanation, or revision.

Philosophical realism, constructivism, pragmatism, and social epistemology offer different interpretations of what a valuation benchmark represents. The formal model does not settle those debates. It instead requires the researcher to state whether V_i^* is intended as a structural economic value, an expected-payoff measure, a welfare measure, a legal claim value, or a normative objective.

Religious and ethical traditions may inform institutional purposes, duties, distributive priorities, and conceptions of welfare. They do not automatically supply empirical estimates of prices, elasticities, or risk. Where ethical commitments enter the model, they should appear transparently in the objective function, constraints, or governance charter rather than being hidden inside a purportedly neutral parameter.

16 Political science, geopolitics, diplomacy, and international relations

Institutions may be embedded in states, central banks, exchanges, treaty organizations, or cross-border regulatory networks. Their valuations can depend on sanctions, capital controls, reserve status, payment infrastructure, treaty obligations, security guarantees, jurisdictional risk, and the credibility of policy commitments.

A geopolitical scenario model can be represented as

$$V_i^* = \sum_{\omega \in \Omega} \Pr(\omega \mid X)\, V_i^*(\omega), \tag{65}$$

where Ω is a set of mutually exclusive scenarios. Scenario probabilities must be justified, calibrated, or explicitly labeled as subjective. They should not be treated as objective frequencies when the events are unique or structurally changing.

Diplomatic reconciliation differs from valuation reconciliation. A negotiated settlement may be valuable because it changes future behavior, lowers conflict risk, or creates a credible commitment. Yet a diplomatic agreement does not by itself prove that the parties' valuations of underlying assets or obligations are accurate. The model should distinguish a valuation disagreement from a distributional conflict, strategic bargaining problem, or disagreement over legitimate authority.

17 Welfare economics and distributional consequences

Minimizing $\mathcal{J}$ is not automatically equivalent to maximizing social welfare. The weights w_i, a_i, b_i, λ encode priorities, and a low disagreement score may conceal unequal burdens or externalities. If valuations influence resource allocation, the analysis should report who gains, who loses, and which constraints protect affected parties.

Let x denote a policy or allocation and $W(x)$ a declared social-welfare criterion. A policy problem might be

$$\max_{x} \quad W(x) \tag{66}$$

$$\text{subject to} \quad \mathcal{J}(\mathbf{z}(x)) \leq \bar{J}, \tag{67}$$

$$x \in \mathcal{F}_{\text{legal}} \cap \mathcal{F}_{\text{resource}}. \tag{68}$$

The disagreement threshold is then a governance constraint, not the definition of welfare. Distributional weights, rights, public goods, externalities, and intergenerational effects must be stated explicitly.

18 Warfare economics, security, and resilience

In security settings, valuations may concern logistics, critical infrastructure, sovereign funding, strategic reserves, or continuity of essential services. The framework can support high-level resilience analysis by comparing institutional estimates, identifying uncertainty, and testing alternative scenarios. It should not be interpreted as a license to treat human lives, civilian welfare, or legal protections as mere monetary terms.

A resilience objective may be written as

$$\min_{\pi} \ \mathbb{E}\left[L_{\text{service}}(\pi, \omega)\right] + \kappa \ \text{CVaR}_{\alpha}\left(L_{\text{service}}(\pi, \omega)\right), \tag{69}$$

where π is a policy, L_{service} is loss of essential service under scenario ω, and CVaR_{α} is conditional value at risk. This is a generic resilience objective. Its parameters and constraints require public-policy, legal, and domain-specific review. The model should prioritize civilian protection, continuity of essential services, accountability, and compliance with applicable humanitarian and security law.

19 Computer science, algorithms, and system architecture

A production implementation should separate data ingestion, validation, benchmark estimation, valuation reconciliation, constraint checking, reporting, and audit logging. A minimal modular architecture is:

1. `data`: validated observations and metadata;
2. `benchmark`: structural estimate and uncertainty distribution for V_i^*;
3. `valuation`: S_i, D_i, error metrics, and disagreement metrics;

4. `constraints`: multiplicative branches, market clearing, and legal limits;
5. `estimation`: objective function and parameter fitting;
6. `validation`: unit tests, sensitivity tests, and out-of-sample evaluation;
7. `audit`: immutable versioned records of inputs, model versions, and outputs;
8. `interface`: human-readable reports and challenge procedures.

19.1 Reference pseudocode

```
INPUT:
    observations Y
    institutions I = {1, 2}
    asset set M
    model specification and legal constraints

VALIDATE timestamps, units, missingness, provenance, and permissions
ESTIMATE structural parameters theta with uncertainty diagnostics
ESTIMATE or update intrinsic benchmarks V_star[i]
LOAD self-valuations S[i] and dual-valuations D[i]

FOR each institution i:
    self_error[i] = S[i] - V_star[i]
    dual_error[i] = D[i] - V_star[i]
    disagreement[i] = S[i] - D[i]
    phi[i] = (S[i] - D[i]) *
             (D[i] - V_star[i]) *
             (V_star[i] - S[i])

CHECK market-clearing residuals and relative-price identities
CHECK legal, accounting, physical, and governance constraints
COMPUTE disagreement loss D_loss
COMPUTE accuracy loss A_loss
COMPUTE joint objective J

IF benchmark identification or data quality is inadequate:
    REPORT uncertainty and do not claim verified reconciliation
ELSE:
    SOLVE constrained reconciliation problem
    TEST constraint preservation and sensitivity
    PRESENT proposed revisions for authorized human review

LOG inputs, versions, assumptions, uncertainty, overrides, and decisions
OUTPUT valuations, disagreement, error estimates, constraint status,
       scenario analysis, and an auditable explanation
```

19.2 Algorithmic safeguards

The multiplicative equality is numerically delicate: floating-point calculations rarely return exact zero. For computation, define a scale-aware tolerance $\epsilon > 0$ and test

$$|\Phi_i| \leq \epsilon \, K_i, \tag{70}$$

where K_i is a scale term with units matching Φ_i, chosen so the test is invariant to sensible rescaling. Because the product has cubic units in valuation, comparing $|\Phi_i|$ with a fixed dimensionless tolerance is invalid unless the variables have first been normalized.

The disjunctive constraint may create a nonconvex feasible set. A solver must either handle the branch structure explicitly or use a justified mixed-integer, complementarity, or branch-enumeration formulation. A local optimizer may find a branch-specific local solution without establishing global optimality.

20 AI, machine learning, and model risk

Machine learning may estimate nonlinear demand, liquidity, network effects, or the mapping from observed features to valuation benchmarks. It should be used within an economic and governance structure rather than as an unexamined replacement for it.

A generic predictive model is

$$\widehat{V}^{*}_{i,t} = f_\theta(X_t), \tag{71}$$

where f_θ may be a regression, tree ensemble, neural network, state-space model, or hybrid architecture. Predictive performance does not by itself establish causal validity, structural identification, or normative legitimacy.

Minimum safeguards include:

1. temporal train–validation–test separation;
2. leakage prevention and point-in-time data reconstruction;
3. benchmark comparisons against simple baselines;
4. calibration and uncertainty intervals;
5. subgroup and regime performance checks;
6. drift monitoring and documented retraining triggers;
7. interpretability proportional to decision impact;
8. human review for consequential or legally regulated decisions;
9. reproducible seeds, dependency versions, and model cards;
10. independent red-team testing and access control.

20.1 Causal inference versus prediction

Let Y be an outcome and A an intervention. A predictive association

$$\mathbb{E}[Y \mid A = a] \tag{72}$$

need not equal the causal estimand

$$\mathbb{E}[Y \mid \mathrm{do}(A = a)]. \tag{73}$$

The distinction matters when a valuation model is used to recommend a policy or attribute the effects of a regulatory, monetary, or geopolitical event. Causal claims require an identification strategy and sensitivity analysis.

21 Cybersecurity, privacy, and operational resilience

Valuation systems may expose commercially sensitive information, market positions, sovereign data, or confidential institutional assessments. Security controls should include least-privilege access, strong authentication, encryption in transit and at rest, key management, signed model artifacts, dependency pinning, patching, backups, incident response, and tested recovery procedures.

A threat model should identify assets, actors, trust boundaries, attack surfaces, and plausible impacts. Important risks include data poisoning, model extraction, unauthorized valuation changes, compromised dependencies, credential theft, manipulation of input timestamps, denial of service, and insider misuse.

The audit log should be tamper-evident, but immutability must be reconciled with lawful correction, retention, and deletion obligations. Sensitive data should be minimized; a system should not collect personal or clinical information merely because psychology or psychiatry appears among the possible interpretive fields. Access to confidential dual valuations should be controlled by mandate and purpose.

22 Cross-institution consistency and network generalization

The two-institution structure can be generalized to n institutions. Let S_i be institution i's self-valuation and $D_{i\leftarrow j}$ be institution j's valuation of institution i. The pairwise disagreement graph has nodes i and directed edges representing assessments. A weighted disagreement objective is

$$\mathcal{D}_n = \sum_{i=1}^{n} \sum_{j \neq i} w_{ij} \left(S_i - D_{i\leftarrow j}\right)^2 . \tag{74}$$

If institutions assess a shared object with a common benchmark V^*, an accuracy term can be

$$\mathcal{A}_n = \sum_{i=1}^{n} a_i (S_i - V^*)^2 + \sum_{i=1}^{n} \sum_{j \neq i} b_{ij} (D_{i\leftarrow j} - V^*)^2 . \tag{75}$$

A network model introduces new issues: missing edges, correlated assessments, common information sources, strategic reporting, and cycles of influence. Agreement propagated through a network is not independent evidence if institutions merely copy one another.

For two institutions, the dual relationship is especially transparent. For larger networks, the source and direction of each assessment must remain explicit to prevent a circular consensus from being mistaken for independent corroboration.

23 Proposed empirical hypotheses

The framework suggests testable hypotheses, not conclusions that follow from its definitions.

1. **H1:** Improvements in independently measured benchmark quality reduce expected absolute valuation error, conditional on stable institutional incentives.
2. **H2:** A fall in $|S_i - D_i|$ does not necessarily predict a fall in benchmark error.
3. **H3:** Explicit value-accounting records improve the attribution of valuation revisions to fundamentals, beliefs, and cross-institution reassessment.
4. **H4:** Market-equilibrium variables improve benchmark estimation only when they contain information not already embedded in the institutional valuations and the model is adequately identified.

5. **H5:** Constraint-aware reconciliation produces fewer infeasible outputs than unconstrained reconciliation, but may be more sensitive to model misspecification.

6. **H6:** Governance procedures that preserve independent review reduce correlated error only when institutions have sufficiently independent evidence or incentives.

A suitable evaluation compares the proposed model with baseline methods: simple averaging, independent institution forecasts, a benchmark-only model, a disagreement-only objective, and a joint objective. Evaluation metrics may include mean absolute error, root mean squared error, calibration, constraint-violation rates, stability, revision attribution, and welfare-relevant outcome measures. The correct metrics depend on the use case.

24 Limitations and failure modes

The principal limitations are structural:

1. **Latent benchmark:** V_i^* may not be directly observable.

2. **Model dependence:** the benchmark can inherit misspecification and normative choices.

3. **Consensus bias:** the institutions may share data, incentives, or errors.

4. **Scale mismatch:** valuation variables may use incompatible units, horizons, or conventions.

5. **Nonconvexity:** the multiplicative constraint is a union of branches.

6. **Strategic manipulation:** institutions may report valuations to influence the other party or the solver.

7. **Market frictions:** observed prices may diverge from frictionless equilibrium.

8. **Legal infeasibility:** an objective-minimizing result may violate binding rules or mandates.

9. **Distributional blindness:** a scalar objective may hide unequal effects.

10. **Overgeneralization:** analogies from physics, biology, psychology, or religion do not automatically transfer causal mechanisms.

A credible implementation should report these limitations alongside estimates and should distinguish a mathematical consistency check from an empirically validated model.

25 Integrated conceptual diagram

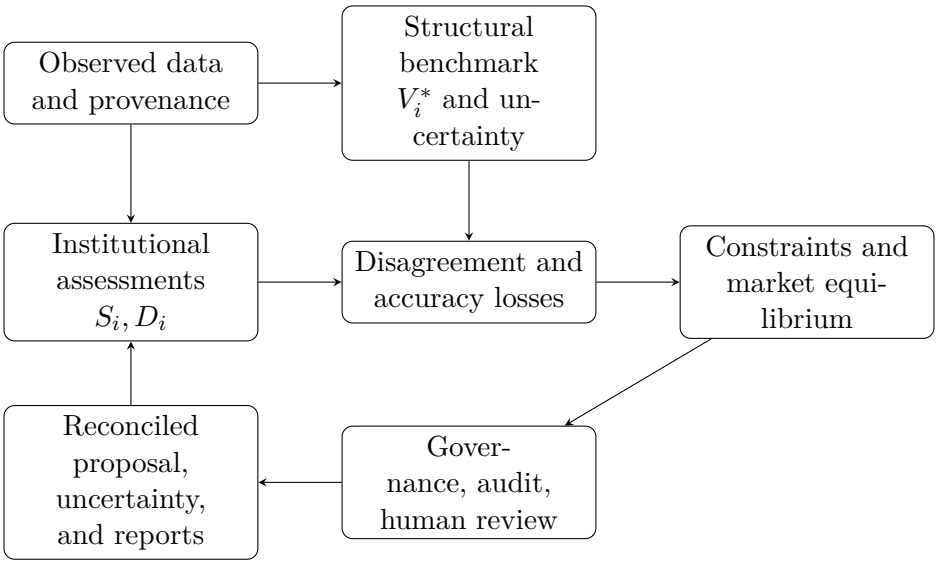


Figure 1: Conceptual workflow. The feedback arrow represents a documented revision cycle, not automatic acceptance of a model output.

26 Conclusions

The Two-Institutions Model formalizes a distinction that is easy to lose in practice: agreement is not accuracy, market price is not automatically intrinsic value, and an institutional judgment is not the same thing as an independently identified benchmark.

The multiplicative condition

$$(S_i - D_i)(D_i - V_i^*)(V_i^* - S_i) = 0 \tag{76}$$

states that at least two of the three valuations coincide. It does not alone select the fully reconciled state. A loss function that rewards accuracy, explicit feasibility conditions, and a defensible benchmark are required to make reconciliation substantively meaningful.

Value-based accounting tracks changes in the benchmark and in institutional assessments separately. Dynamic adjustment can reduce disagreement under specified stability conditions, but disagreement convergence is not equivalent to truth convergence. Monetary-asset demand and supply provide useful external restrictions, especially through relative prices, liquidity, scarcity, network effects, institutional credibility, and equilibrium conditions. Yet market clearing does not uniquely identify intrinsic value without further structural assumptions and evidence.

The most defensible implementation is therefore modular, auditable, uncertainty-aware, legally constrained, and open to challenge. Mathematics specifies relationships; economics supplies behavioral and equilibrium structure; econometrics addresses identification; computer science provides reproducible procedures; governance determines authority and accountability; and ethical analysis examines distributional consequences. No single discipline substitutes for all the others.

The framework's central operating principle is consequently:

$$\boxed{\text{Reduce disagreement without making consensus the definition of truth.}} \tag{77}$$

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Glossary

Accuracy loss
A loss function measuring distance between assessments and an explicitly defined benchmark.

Benchmark
The reference value against which assessments are evaluated; it may be estimated and uncertain.

Constraint preservation
A procedure that keeps numerical updates within a specified feasible set.

Dual valuation
The valuation assigned to an institution's object or position by the other institution.

Intrinsic-value benchmark V_i^*
A model-defined or independently measured reference value not defined solely by institutional consensus.

Market clearing
Equality of quantity demanded and quantity supplied under the specified market model.

Multiplicative condition
The product of the three pairwise valuation differences equals zero, implying at least one pair is equal.

Relative price
The price of one asset expressed in units of another asset.

Self-valuation S_i
An institution's own assessment.

Structural identification
Conditions under which the data and model restrictions distinguish the target parameter or benchmark from alternatives.

Value accounting
The separate recording of changes in the benchmark, self-valuation, dual valuation, errors, and disagreement.

Valuation disagreement Δ_i
The signed difference $S_i - D_i$.

The End